Course Title: CS 2216

Course Name: Principles of Computer and Network Security

Number of Credits: 10

Duration: 15 weeks of classes

Number of contact hours: 3 lecture hours or its equivalent per week

Prerequisites: CS 2103 – Data Communication and Computer networks

Course offered by: Department of Computer Science and IT

Course description:
This course provides an overview of the major topics in computer and network security based on 10 major domains of Common Body of Knowledge (CBK) defined by the International Information Systems Security Certifications Consortium. Students will be exposed to information and organizational security, risk management, disaster recovery, cryptography and public key infrastructure, network and infrastructure security, and security baselines. A part from that, topics cover attacks and malware, email security and web components, access control and privilege management, computer forensic, ethics and laws related to information security.

Learning outcomes:
At the end of the course, students should be able to:
1. Demonstrate the understanding on the importance of information security awareness to be able to appreciate IT security and control practices
2. Demonstrate the understanding on the overview of information security concepts and principles,
3. Identify threats and vulnerabilities in information systems,
4. Practice secure control while using computing and networking system
5. Use and demonstrate security tools given in the course.

Course outline:

Week 1: Information Security Overview
- What is information security?
- Security trends, confidentiality, integrity, availability and other security concepts.
- Security Incidents
- Why Security?
- Growing IT Security Importance and New Career Opportunities
- Twelve (12) Information Security Principles

Week 2: Organisational / Operational Security
- Security operations in organization
- Physical security safeguard
- The role of people in security

Week 3: Risk Management
- Overview of Risk Management
- Risk management models
- Qualitative and quantitative risk assessment

Week 4: Disaster Recovery, Business Continuity, and Organisational Policies

- Overview of the Organizational Policies
- Disaster Recovery Planning
- Business Continuity Plan

Week 5-6: Cryptography & Public Key Infrastructure

- Algorithm & Hashing
- Symmetric
- Asymmetric
- Digital Certificates

Week 6-7: Network & Infrastructure Security

- OSI Model, Network Protocol
- Remote Access
- Intrusion Detection System

Week 8-9: Security Baselines

- Password Selection
- OS Hardening
- Network Hardening
- Application Hardening

Practical 1: Practical practice of how to perform hardening process on Windows platform.

Week 10-11: Attacks and Malware

- Denial of Service
- Sniffing/spoofing/social engineering
- Wardialling&Wardriving
- Malware

Week 11-12: Email Security and Web Components

- Malicious codes & hoax email
- Spamming
- Mail encryption Protocols
- Code-based vulnerabilities
- Symmetric cryptographic

Week 13-14: Access Control and Privilege Management

- Identification & authentication techniques
- User, group & role management
- Handling access control

Week 15: Introduction to Computer Forensics, Ethics and Laws

- Digital Evidence
- Collection and preservation of evidence
- Free space vs slack space
- Analysis and investigation
- Computer crime categories
- Privacy laws

Method of Instruction:

Mix of lectures, discussions and lab practical

Assessment methods:

The course assessment is based on individual and group assignments, and tests. The course work weighs 40% and final exam 60%. The course work comprises group project assignment (15%), mid-semester test (15%) and one more written test (10%). Final marks graded from A to C as pass, D and E as fail.

Text Books:

1. Conklin, W.A., White, G.B., Cothren, C., Williams, D., & Davis, R.L. (2005). *Principles of computer security: Security+ and beyond.* Singapore: McGraw Hill.

References:

1. Anderson, R. J. (2001). *Security engineering: A guide to building dependable distributed systems*. New York: John Wiley.
2. Gollman, D. (2003). *Computer security*. West Sussex: John Wiley.
3. Krause, M., & Tipton, H.F. (2001). *Handbook of information security management.* Canada: CRC Press.
4. Merkow, M., &Breithaupt, J. (2006). *Information security principles and practices.* New Jersey: Pearson Prentice Hall.
5. Pfleeger, C.P., &Pfleeger, S. L. (2003). *Security in Computing.* New. Jersey: Pearson Education Inc.